OSI Model (Open Systems Interconnection Model)

1. Introduction

The Open Systems Interconnection (OSI) model describes seven layers that computer systems use to communicate over a network. The OSI model is divided into seven distinct layers, each with specific responsibilities, ranging from physical hardware connections to high-level application interactions.

Each layer of the OSI model interacts with the layer directly above and below it, encapsulating and transmitting data in a structured manner. This approach helps network professionals troubleshoot issues, as problems can be isolated to a specific layer. The OSI model serves as a universal language for networking, providing a common ground for different systems to communicate effectively.

The OSI model was the first standard model for network communications, adopted by all major computer and telecommunication companies in the early 1980s. It was introduced in 1983 by representatives of the major computer and telecom companies, and was adopted by ISO as an international standard in 1984.

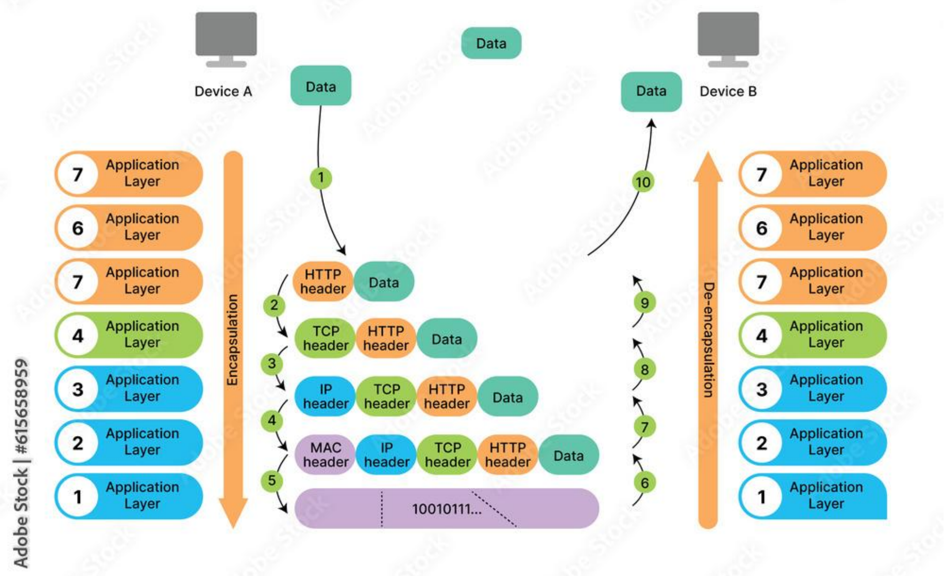
2. Working of the OSI model

As technologies evolve and new models emerge , the OSI model remains integral to understanding metwork architecture. Whether a team is designing a simple local area network(LAN) or managing a complex global network, the principles of the OSI model provide a clear, structured approach to networking.

The OSI model comprises of total seven distinct layers. The application layer (layer 7), the presentation layer (layer 6) and the session layer (layer 5) comprise the software layers of an OSI, where all the transmissions occur.

The transport layer (layer 4) is the “heart of OSI,” handling all data communication between network and systems. And lastly, the network layer (layer 3), the data layer (layer 2) and the physical layer (layer 1) controls how the data moves through the physical components of the network.

Data moves bi-directionally through the OSI model. Here each layer communicates with the layers below and above in the stack. Both the sending and receiving devices transmit data through the data layers.

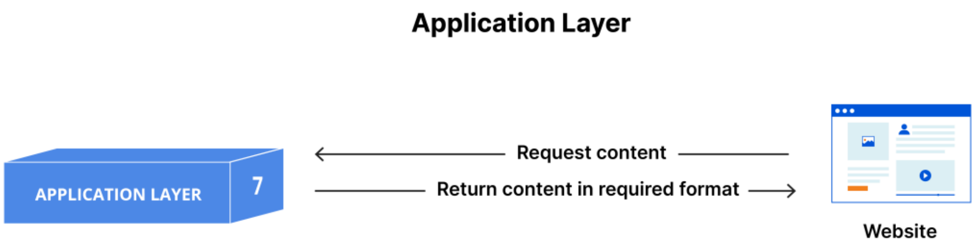


Seven Layers of the OSI model

The OSI model acts as a foundation for protocol development, with each layer of the framework managing specific network processes.

Layer 7: The application layer

The application layer is the OSI layer closest to the end user. It provides network services directly to user applications and facilitates communication between API endpoints and lower layers of the OSI model.



Applications themselves are not part of this layer. Rather, the application layer provides the protocols (HTTP, FTP, DNS etc) that enable software to send and receive data. It's responsible for processes such as:

- **File transfer -**The application layer takes human-readable data files from the user's device and transmits them to the presentation layer.
- **Communication and authentication -** The application layer makes sure that the receiving device can accept the data and that the communications interfaces needed for the transfer exist. It can also be used to authenticate the devices involved in the transfer.

Layer 6: The presentation layer

The presentation layer transforms data into a format that the application layer can accept for transmission across the network. Due to its role in converting data and graphics into a displayable format for the application layer, it is sometimes referred to as the syntax layer.

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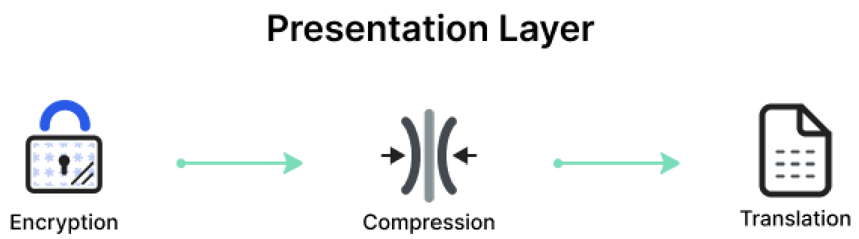
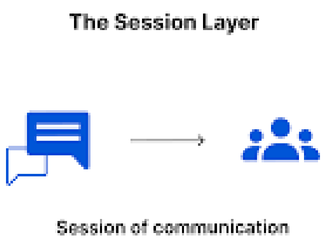


The presentation layer is responsible for:

- **Data translation -** The presentation layer converts data into the correct format during the encapsulation process.
- **Data encryption and decryption -** The presentation layer encrypts data for secure transmission and decrypts it upon delivery.
- **Data compression** - The presentation layer reduces the size of a data stream for transmissions and decompresses it for use.

Layer 5: The session layer

The session layer is the process of establishing, managing and terminating connections called "sessions" between two or more computers. It initiates the connections between local and remote applications, keeping the session open.



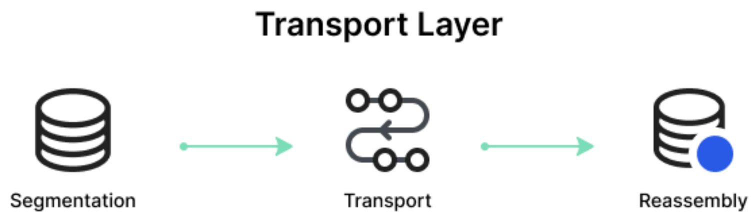
Key functions of the session layer include:

- **Session interactions -** The session layer manages user log on (establishment) and user log off (termination), including any authentication protocols integrated into client software.

- **Synchronization -** The session layer helps ensure that data streams are properly synchronized and handles recovery points.

Layer 4: The Transport layer

The transport layer uses protocols like transmission control protocol (TCP) and the user datagram protocol (UDP) to manage the end-to-end delivery of complete messages. It takes messages from the session layer and breaks them into smaller units (called "segments").



The transport layer also handles:

- **Service point addressing -** The transport layer helps ensure that messages are delivered to the correct process by attaching a transport layer header (including a service point or port address).
- **Flow control -** The transport layer prevents data overflow and manages the rate of data transmission between two devices interacting on the network.

Layer 3: The Network layer

The network layer of the OSI model is responsible for facilitating data transfer from one node to another across different networks. The network layer determines the best path (routing) for data to travel between nodes. If segments are too large, the network layer breaks them up into smaller "packets" for transport and reassembles them on the receiving end.

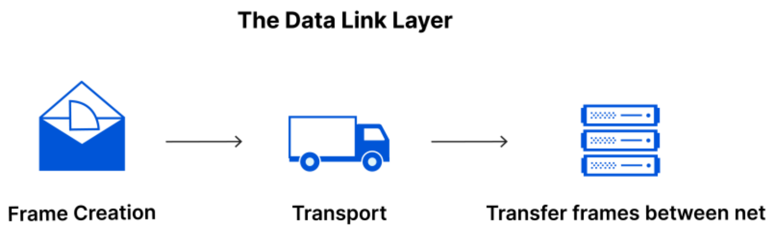
The network layer primarily uses the Internet Protocol v4 (IPv4) and IPv6 and is responsible for:

- **Packet fragmentation and reassembly -** The network layer splits large packets (those that exceed the size limits of the data link layer) into smaller ones for transmission and reassembles them at the destination.

- **Traffic control -** The network layer manages network traffic to prevent congestion and safeguard efficient data flow.

Layer 2: The Data Link layer

The data link layer's primary function is to manage error-free data transfer between multiple devices interacting on the same network.



The DLL is divided into two sublayers.

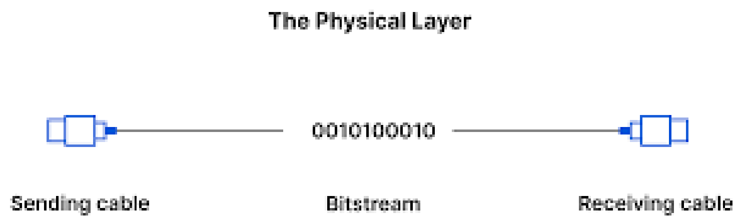
The **logical link control (LLC) layer** serves as an interface between the media access control (MAC) layer and the network layer-handles flow control.The **MAC layer** controls how devices access network mediums and transmit data.

DLL functions include:

- **Framing -** The DLL allows the sender to transmit a set of bits (data) that are meaningful to the receiver by attaching special bit patterns to the beginning and end of the frame.

Layer 1: The Physical layer

The physical layer comprises the physical network components responsible for transmitting raw data-in the form of "bits" or strings of 1s and 0s-between devices (connectors, routers, repeaters and fibre optic cables, for instance) and a physical medium (like wi-fi).



The physical layer is responsible for:

- **Bit rate control -** The physical layer defines the data transmission rates, often in bits per second.
- **Bit synchronization -** The physical layer imposes a clock on bit streams, ensuring that the sender and receiver are synchronized at the bit level.
- **Physical topologies -** The physical level specifies how network devices and nodes are situated. Standards like USB, Bluetooth and Ethernet include physical layer specifications.

3. Benefits of the OSI model

Accelerated research and development

The modularity of the OSI model encourages faster innovation in protocol development, since software engineers can integrate new technologies without overhauling the entire network stack.

Streamlined knowledge sharing

Software engineers can separate the operating layers of each network component and organize them according to their primary roles in the network.

Simplified troubleshooting

When a device on the network fails or an app loses connection, the OSI model allows teams to pinpoint and isolate the problem layer to address any security issues or networking vulnerabilities without disrupting the entire framework.